Electricity and Magnetism

Unit Test Practice Worksheet

Name:		Class:		Date:	

Section A: Multiple Choice

Circle the letter of the best answer for each question.

1. Which of the following materials is commonly used to make permanent magnets?
   A) Plastic
   B) Rubber
   C) Iron
   D) Glass

2. What determines whether or not a material is magnetic?
   A) The density of the material
   B) The weight of the material on Earth
   C) The alignment of atoms in the material
   D) The number of elements that make up the material

3. A repelling force occurs between two charged objects when the charges are of:
   A) Like signs
   B) Unlike signs
   C) Equal magnitude
   D) Unequal magnitude

4. When a rubber rod is rubbed with fur and gains a negative charge, what happens?
   A) Negative charges move from the rod to the fur
   B) Negative charges move from the fur to the rod
   C) Negative charges are added to the rod, and the fur stays neutral
   D) Positive charges are removed from the rod, and the fur stays neutral

5. What happens to a charged object placed in an electric field?
   A) It always moves away from the field
   B) It only moves if it is heavier than the field
   C) It is attracted or repelled depending on its charge
   D) It is unaffected by the field

6. On a bar magnet diagram, where is the magnetic force the strongest?
   A) At the poles (ends) of the magnet
   B) To the side of the magnet

C) Far away from the magnet
D) The magnetic force is equal at all points

7. Which of the following best describes a gravitational field?
A) It is the space around a charged object
B) It affects only objects in direct contact
C) It surrounds a mass and pulls other masses toward it
D) It repels objects away from the source mass

8. What is the best definition of electromagnetism?
A) The use of electrical energy to produce magnets
B) The use of magnets to generate electric current
C) The interaction between electricity and magnetism
D) Objects in nature that have both electric and magnetic fields

9. How would increasing the electrical current affect an electromagnet?
A) It would make the magnetic field weaker
B) It would make the magnetic field stronger
C) It would decrease the flow of electricity in the iron bar
D) It would increase the flow of electricity in the iron bar

10. Which statement describes how electromagnetism is used in an electric generator?
A) Moving a magnet in a coil of wire produces visible light
B) Magnets convert electric energy into mechanical energy
C) Magnets detect the presence of electric currents in nature
D) Moving a magnet in a coil of wire generates an electric current

Section B: Calculations

Use Ohm's Law: **V = IR I = V/R R = V/I** *(V = volts, I = amperes, R = ohms)*

Show all work and include units in your answers.

11. A resistor has a resistance of 20 Ω and a current of 4 A flowing through it. Calculate the voltage across the resistor.

Given: R = _____ I = _____ V = ?
Formula: _____
Work:

Answer: V = _____

12. A lamp operates with a voltage of 120 V and a current of 2 A. Determine the resistance of the lamp.

Given: V = _____ I = _____ R = ?
Formula: _____
Work:

Answer: R = _____

13. A parallel circuit has a total voltage of 36 V and three resistors: 10 Ω, 20 Ω, and 30 Ω. Find the total resistance.

(Hint: $1/R_{total} = 1/R_1 + 1/R_2 + 1/R_3$)

Given: R1 = _____ R2 = _____ R3 = _____
Work:

Answer: R_{total} = _____

14. A circuit has a 60 V battery and two resistors (5 Ω and 10 Ω) connected in series. Calculate the total current.

(Hint: In series, $R_{total} = R_1 + R_2$)

Given: V = _____ R1 = _____ R2 = _____
Work:

Answer: I = _____

Section C: Short Answer

Answer in complete sentences. Use key vocabulary where appropriate.

15. If the charge on an object is greater, what happens to the strength of the electric force needed to move it? Explain.

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16. Draw a schematic diagram of a series circuit that includes: one 9V battery, 3 lamps, and one switch. Label each component.

Drawing space:

17. Explain what happens in a coil of copper wire when a magnet begins to spin inside it (as in a wind turbine generator).

18. What is the name of the process in which a spinning magnet inside a coil of copper wire generates electricity?

19. Explain how a wind turbine produces electricity. Describe the step-by-step process and the forms of energy that are converted.

20. Explain the environmental impact of generating electricity from solar energy. Include both positive and negative effects.